

lies and says No (Case 4)

[Note:] This question needs further study. Please read the Book's written by Raymond Smullyan

20) An explorer is captured by a group of cannibals. There are two types of cannibals—those who always tell the truth and those who always lie. The cannibals will barbecue the explorer unless he can determine whether a particular cannibal always lies or always tells the truth. He is allowed to ask the cannibal exactly one question.

- a) Explain why the question "Are you a liar?" does not work.
- b) Find a question that the explorer can use to determine whether the cannibal always lies or always tells the truth.

[Soln:] a) Suppose, the explorer asks "Are you a liar?" The cannibal replies—~~Yes~~ If the cannibal is a truth speaker, he will say No, because he is not a liar. If the cannibal is a liar, he must lie. So he can't say Yes. He too must speak 'No'. So, we have no way of differentiating between a truth-speaking cannibal and a liar cannibal. $\therefore$ The question doesn't work.

b) **[Claim:]** The question "Do you eat human flesh?" will work.

[Proof:] If the cannibal is a truth speaker, he will say Yes. If the cannibal is a liar, the answer will be No. All cannibals, by defn, eat human flesh.

21) When three professors are seated in a restaurant, the hostess asks them: "Does everyone want coffee?" The first professor says "I do not know." The second professor then says "I do not know." Finally, the third professor says "No, not everyone wants coffee." The hostess comes back and gives coffee to the professors who want it. How did she figure out who wanted coffee?

[Soln:] If the first professor would not want coffee, he would have answered No, since he will not be included. Since he does not know what the other professors want and given his answer, we imply he wants coffee.